

Predicting housing insecurity for families

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Where I come from: mathematics, and building methods

Training

- PhD in Mathematics, Shahid Beheshti University, 2024. Graduated first in the department
- MSc in Soft Computing, first in class; BSc in Mathematics, first in class
- Doctoral work on algebraic structures in non-classical logic: hoops, lattice implication algebras, filters and ideals
- Postdoctoral Fellow, Signal and Image Processing Laboratory, Shiraz University, on the National Elite Research Scheme

Output

- Seven peer-reviewed journal articles and one conference paper
- One first-author manuscript under review at *Expert Systems With Applications*

What the mathematics is for

- I read and write formal notation, so I can check whether a model's assumptions actually hold rather than take them on trust
- Proving results in algebra is the same discipline as auditing an estimator: state the assumptions, then find the case that breaks them
- **I design methods, not only apply them.** My current manuscript proposes a factorised invariant and equivariant extension of Barlow Twins, a self-supervised objective that separates what a representation should keep constant from what it should let vary
- That habit is why this project reports where the model is unnecessary as carefully as where it helps

Machine learning in practice

Architectures I work with

- Transformers and attention, including BERT-style language model encoders for text, and vision and multimodal variants
- Autoencoders and variational autoencoders
- Self-supervised representation learning, including my own extension of Barlow Twins
- Bayesian deep learning with Monte-Carlo dropout, which I have also used in this project
- Convolutional and fully connected networks, gradient boosting, random forests, penalised regression
- Survival models, clustering, causal inference with inverse-probability weighting

Where I have used them

- **Bayesian deep survival modelling**, METABRIC, 1,979 patients. Monte-Carlo dropout, leak-free cross-validation, D-calibration, selective prediction and SHAP
- **Pan-cancer classification**, 801 samples across 20,531 genes, with a browser-based dashboard that predicts live
- **Multimodal fusion**, 47,681 spatial transcriptomic spots aligned to paired histology images
- **Survival-aware multi-omic clustering**, 1,860 patients, with inverse-probability-weighted Cox models
- **Semi-self-supervised segmentation**, published in *Plant Phenomics*

PyTorch, scikit-learn, R. Docker, Snakemake, Nextflow and Slurm HPC. Code released with documentation.

Coming to this from a different field

Where this is new for me

- My applied work so far has been clinical, genomic and imaging rather than housing
- Local authority and social care data would be a new setting
- Homelessness policy is a field I have been reading into rather than working in

The methods travel well between domains. Rather than ask you to take that on trust, I tested it.

So I ran the experiment

- Last week I worked through this project's first month on England's published homelessness returns

What I built (all public, Open Government Licence)

- MHCLG statutory homelessness detailed local authority tables, 2018–19 to 2024–25, joined to the 2019 deprivation indices
- 2,167 authority-years, 83 features
- **Question:** which areas will see families with dependent children become homeless next year, and can we know beforehand?
- **Discipline:** a strictly prospective backtest. Train on the past, score a year the model has never seen, and fix every threshold from the training years alone

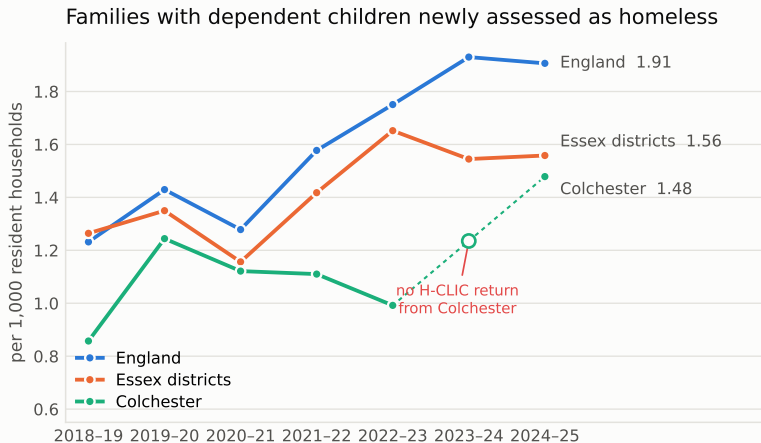
What carries over

Published statistics are aggregate, so this predicts areas rather than households. What transfers is the method, the validation design and the failure modes I ran into, which is what I would bring to CBH's household-level data on day one.

The machine learning, and what each part is for

	Task	Method	Held-out result
Supervised	Level. Worst national quintile next year?	Penalised logistic regression, gradient boosting, isotonic calibration, MC-dropout network	ROC-AUC 0.95 a sorted column gets 0.90
	Escalation. A rise of 25% or more next year?	The same four, fitted separately	ROC-AUC 0.61 the heuristic gets 0.35
Unsupervised	Segmentation. Which neighbourhoods behave alike?	<i>k</i> -means on nine deprivation domains, <i>k</i> chosen by silhouette	5 service profiles across 105 LSOAs
	Anomaly detection. Which are unusual for Colchester?	Isolation Forest, 500 trees	11 flagged, 5 missed by a deprivation ranking
Supporting	Calibration. Does the score mean what it says?	Isotonic regression on held-out folds	Brier 0.19 to 0.089
	Uncertainty. How sure is the model here?	Monte-Carlo dropout, 100 stochastic passes	A spread per authority, not just a score
	Interpretation. What is driving the score?	Standardised coefficients, direction and magnitude	A ranking an officer can disagree with

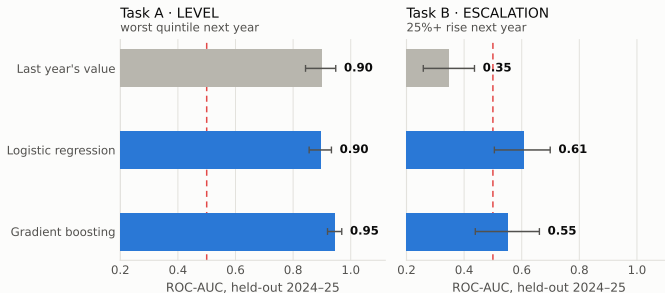
Finding 1: the trend, and a gap in the record



Across England the rate went from **1.23 to 1.91** per 1,000 households. In Colchester the number of families with children owed a relief duty rose from 69 to 126, an increase of **83%**.

Colchester filed no H-CLIC return in 2023-24, one of 18 authorities that year. Sum the missing values to zero and the chart shows a collapse in family homelessness that never happened.

Finding 2: where prediction actually earns its place

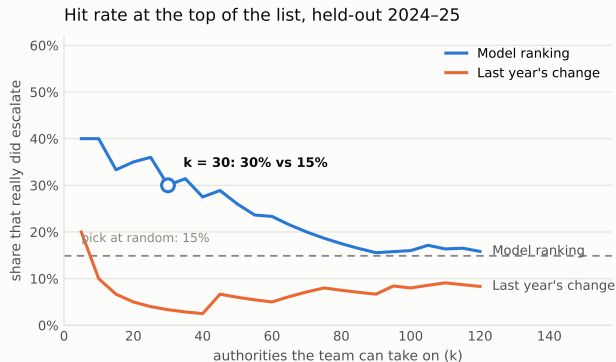


Dashed line marks chance (0.5); bars carry 95% bootstrap intervals. Last year's value already answers Task A; on Task B it is significantly worse than chance.

Task A is a lookup. Last year's value scores 0.90. A model reaches 0.95, which is worth having for the calibration but not for the ranking. **There is no point building machine learning to answer a question that a sorted column already answers.**

Task B is the harder one. The obvious heuristic, "it went up last year so watch it", scores 0.35 [0.26–0.44], **significantly worse than chance**. Rises tend to mean-revert, so acting on last year's increase sends help to the wrong places.

Finding 3: what that is worth to a team with a caseload



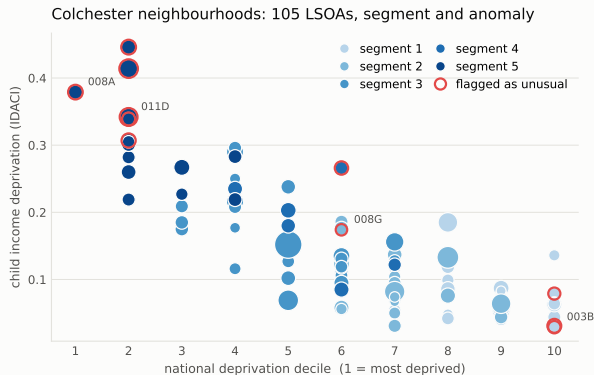
A service cannot act on a ranking of the whole country. It works a list of fixed length, so the metric that matters is the **hit rate at the top of that list** rather than AUC.

At $k = 30$, **30%** of the flagged areas really did escalate, against a base rate of 15%. That is **twice the yield** for the same effort.

The heuristic does worse than picking at random.

Held-out year 2024-25, $n = 262$. The absolute numbers are modest. I would rather show you an honest 2× than a polished 0.9.

Finding 4: the neighbourhood layer, and who a headline misses



Marker area is the number of children aged 0–15. Segments are ordered by child income deprivation.

Colchester has 105 neighbourhoods and 34,370 children aged 0–15. I grouped them into 5 service profiles and used an Isolation Forest to flag 11 as unusual for Colchester.

5 of those 11 sit in the less deprived half of the country and between them contain 1,337 children. A triage that sorts on headline deprivation will not see them.

Risk and deprivation are correlated but they are not the same thing, and the gap between them is where earlier intervention becomes possible.

Built so that someone else can run it

Two layers

- 12 library modules hold the logic: parsing, measure selection, boundary reconciliation, features, the models and their evaluation
- 9 stage scripts run them in order. Each reads one artefact, calls into the library, and writes the next
- Paths, years, split and thresholds are defined once, so the modelling stage and the dataset card cannot disagree about what the test year is

Why it is worth the effort

- 78 tests. The ones that matter most are on the label construction and the metrics, because that is where a bug changes a published number without changing anything visible
- Model configurations are read off the fitted pipelines, so the dashboard cannot describe a model nobody ran
- One command rebuilds the data, the models, the figures, this deck and the dashboard

A research post leaves something behind. This is the shape I would want that to be in.

What this rehearsal is not

Limits I would state in any report

- The published statistics are aggregate, so this predicts areas rather than households. It is a test of method, not a finding about Colchester's families.
- Deprivation is fixed at 2019 while outcomes run to 2025.
- 18 authorities did not report in 2024-25, and non-reporting is almost certainly not random.
- Effect sizes are modest and the intervals are wide.

What changes with CBH's own data

- Household level and monthly rather than annual, so the horizon becomes weeks. That is where prevention actually works
- Rent account trajectories, repairs, anti-social behaviour, contact history and tenancy events, all of which H-CLIC records only after the fact
- Free-text case notes, where transformer-based feature extraction genuinely earns its place
- Children's outcomes through ECC, such as school moves and attendance

The first three months

Month 1

Access and understanding

- DPIA, sharing agreement, lawful basis
- Shadow officers; map how decisions get made
- Agree the outcome, and who acts on it

Month 2

The analysis asset

- Link the data, with lineage and quality flags
- Map non-reporting and suppression
- Profile entries to temporary accommodation

Month 3

Baseline and pilot design

- An interpretable baseline, validated prospectively
- Precision at real caseworker capacity
- Calibration and subgroup equity checks

WHAT EXISTS AT THE END

A governance route and an agreed outcome definition

A documented, reproducible dataset with a data dictionary

A practitioner-reviewed baseline and an evaluation design

Four risks I would manage from day one

Lead time

A correct prediction that arrives in the week a family presents as homeless is worth nothing. The evaluation has to score how early the warning came, not only whether it was right.

Feedback loops

Once officers act on the score, the outcome data is contaminated, because a case that was successfully prevented looks exactly like a false positive. The evaluation design has to be agreed before the pilot starts.

Equity

A model trained on who was served will learn who tends to get served. Subgroup calibration needs to be a standing check rather than a one-off fairness audit.

Proportionality

This is data about children. The test is not whether the model is accurate but whether a family ends up better off. That argues for a model an officer can interrogate, with a human decision at the end of it.

What I would ask in week one

To Colchester Borough Homes

1. What would the team do differently on a Monday morning if a household were flagged? If that is not yet settled, it is the first thing to design together.
2. How many households can you realistically contact in a week? That number defines the metric.
3. Which signals do experienced officers already trust? Those are the features worth starting from.

To the University and ECC

1. What is the route to linking children's education and social care data, and how long does it take?
2. What is already agreed on information governance, and what still has to be negotiated?
3. Which publications and funding bids are we aiming at, so that the analysis can be designed for them from the start?

Thank you. I would be glad to take questions.

Appendix: the pipeline

Stage	Script	What it does
00	00_download.py	Fetch H-CLIC and IoD2019 from gov.uk; checksum each file
01	01_extract_hclic.py	Parse seven publication spreadsheets with stacked merged headers, footnote markers and interleaved subtotals
02	02_build_panel.py	Select measures by published header text, reconcile the 2019–2023 boundary changes and convert counts to rates
03	03_colchester_lsoa.py	Segment and anomaly-detect Colchester's neighbourhoods
04	04_model.py	Prospective backtest, both tasks, bootstrap intervals
05	05_figures.py	Every figure in this deck
06	06_facts.py	Every number in this deck, written out as $\LaTeX$ macros
07	07_dashboard_data.py	The JSON behind the public dashboard
08	08_data_profile.py	The dataset card, measured from the built artefacts

Most stages are under a hundred lines of wiring; the logic they call sits in the library. One command reproduces all of it, and no number here was typed by hand.

Appendix: the library the stages call

Module	What it owns
config	Every path, year, split and threshold, defined once
sources	The gov.uk URLs, checked against the configured years
hclic	The publication-spreadsheet parser
panel	Measure selection, boundary maps, rate conversion
features	Design matrix and the two task labels
models	The model zoo, and what each member is for

Module	What it owns
mlp	The Monte-Carlo dropout network
evaluation	The backtest, the metrics, the bootstrap
neighbourhoods	Segmentation and anomaly detection
plotting	Design tokens and figure plumbing
profiling	The dataset card, measured from the artefacts
export	JSON for the dashboard, macros for these slides

Installed as a package, so the stages, the tests and a notebook all import the same code. Splitting it this way is what makes the parts where a bug would be invisible directly testable.

Appendix: what the tests actually check

Where a silent bug would hurt most

- A label taken from the wrong year, or from a neighbouring authority after a grouped shift
- A quantile cutoff recomputed on the year being scored, which would fix its prevalence at the nominal quintile by construction
- A relative rise computed from a zero baseline
- A missing return arriving as a zero

Each of those has a test whose answer can be worked out by hand.

And the fragile inputs

- The spreadsheet parser, against a synthetic sheet carrying every awkward feature of the real ones: a banner title, stacked merged headers, a percentage sharing a header with its count, region subtotals above the authorities
- Measure selection: the shortest matching header wins, a measure stays inside its own table, an unresolved measure becomes null rather than zero
- The boundary maps: no district claimed by two successors, and none both created and retired

Appendix: boundary and coverage reconciliation

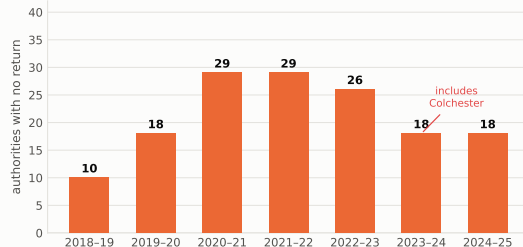
Local authorities move. There were 326 districts in 2018–19 and 296 in 2024–25, while IoD2019 is published on 2019 boundaries.

- Seven successor authorities (Buckinghamshire, the two Northamptonshires, the two Cumbrias, North Yorkshire and Somerset) rebuilt by population-weighting their predecessors' LSOAs
- Fourteen districts abolished in April 2019 mapped onto their successors
- Every authority-year then carries a deprivation score

Measures move too. 251 of the 252 measure-years resolve by matching published header text rather than column position, so a layout change raises an error instead of quietly shifting a feature.

The one gap is real: Section 21 notices were not collected in 2018–19.

Missing H-CLIC returns, by year (out of ~309 authorities)



Appendix: the models that predict

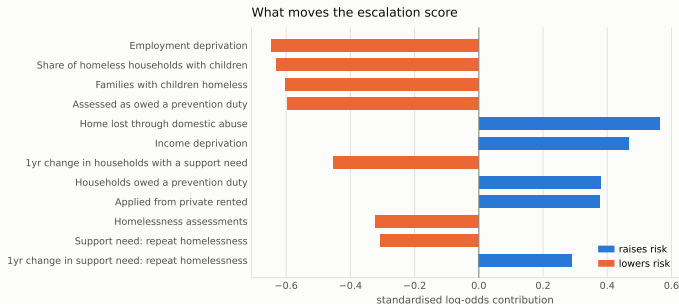
Model	Configuration	Why it is there
Logistic regression	Median impute, standardise, L2 at $C = 0.3$	The interpretable reference and the source of the driver chart. Strongest of the four on escalation
Gradient boosting	Histogram-based, depth 3, 300 iterations, $\eta = 0.06$	Handles the 14% missingness natively rather than imputing values that were never collected
+ calibration	Isotonic, 3-fold	A triage score is read as a probability, so it has to be one. Brier 0.19 to 0.089
Neural network	PyTorch MLP, 64-32, dropout 0.3, reweighted loss, early stopping, Monte-Carlo dropout at inference	Competitive without winning: 0.93 and 0.59, both inside the best model's interval. It is here for the uncertainty, since each authority gets a spread rather than only a score

Each is fitted twice, once per task. No stacking: the differences sit inside their confidence intervals.

Appendix: the baselines, and the neighbourhood models

Model	Configuration	Why it is there
Persistence	Last year's value, or last year's change	What a service already has without any analytics. It answers Task A, and on Task B it is significantly worse than chance. Two of my three findings come from this row
Prevalence	Predict the base rate for everyone	Fixes the floor, so that "30% of the flagged areas" has something to be compared against
<i>k</i> -means	Nine standardised deprivation domains, $k = 5$ by silhouette, restricted to $k \geq 4$	Neighbourhood service profiles. $k = 2$ scores higher but only recovers deprived and not deprived, which the service already knows
Isolation Forest	500 trees, 10% contamination, same nine features	Finds neighbourhoods unusual <i>for Colchester</i> . Contamination is fixed rather than tuned: with 105 points and no ground truth there is nothing honest to tune against

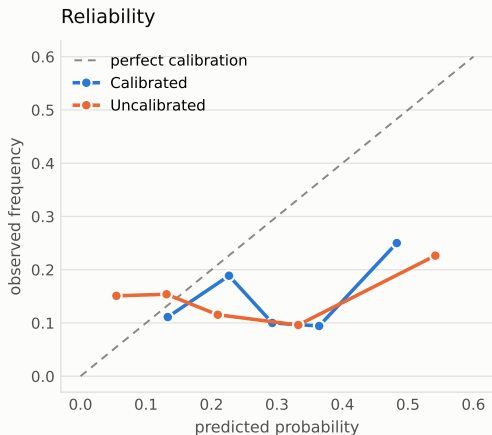
Appendix: what the model keys on



Domestic abuse as a route out of the last settled home **raises** escalation risk. Areas that are already high on family homelessness **lower** it, which is the same mean reversion that makes the persistence heuristic anti-predictive.

The directions are stable and readable, which is what an officer needs in order to disagree with the model.

Appendix: does the score mean what it says?



Whoever acts on a triage score will read it as a probability, so it had better be one. Isotonic calibration takes the Brier score from 0.19 to 0.089 on the level task.

A score that ranks well but is badly calibrated misleads a service about how many families are actually at risk, and therefore about how much capacity to commit. **That is the part I would not compromise on.**

Selective prediction follows the same reasoning. A model that abstains where it is uncertain is more useful than one that always answers.

Appendix: sources

- MHCLG, *Statutory homelessness: detailed local authority-level tables*, financial years 2018–19 to 2024–25. gov.uk/government/statistical-data-sets/live-tables-on-homelessness
- MHCLG, *English Indices of Deprivation 2019*. File 7 (LSOA scores, ranks, deciles and population denominators) and File 10 (lower-tier local authority summaries). gov.uk/government/statistics/english-indices-of-deprivation-2019
- ONS, register of local government reorganisations 2020–2023, for the boundary crosswalk.

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